

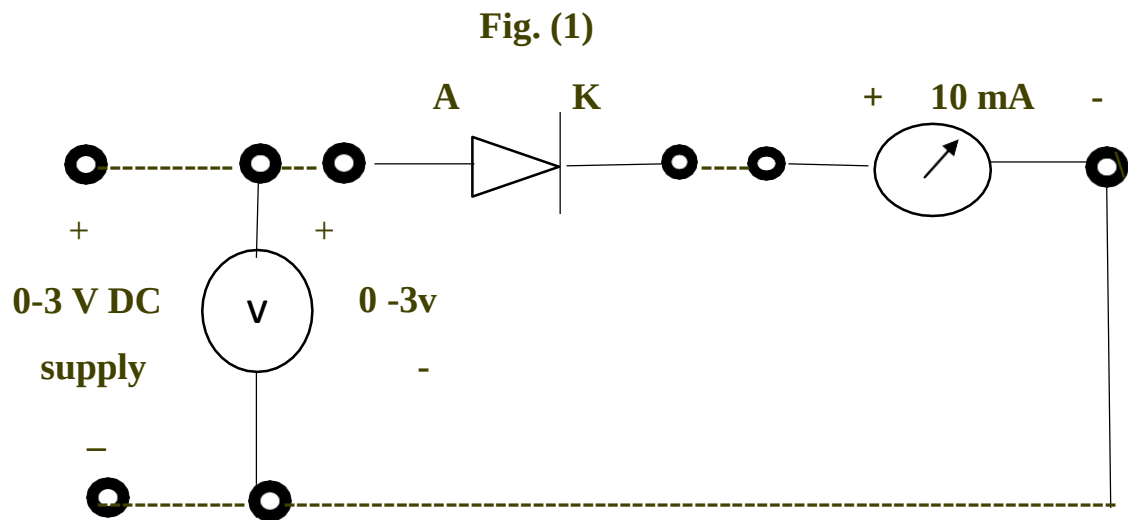
Experiment No. 8

CHARACTERISTICS CURVE OF PN JUNCTION DIODE

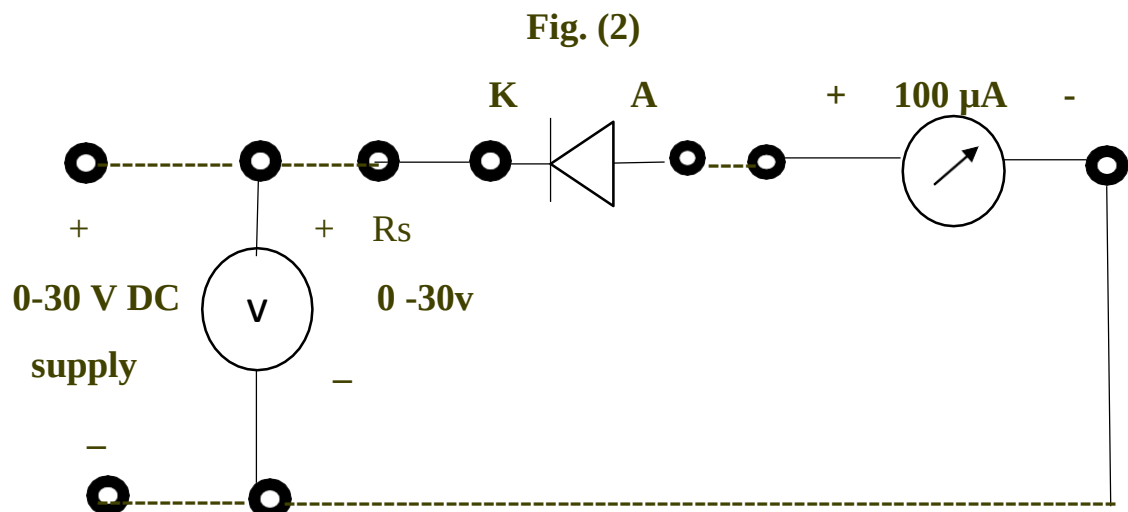
Aim: to study the forward and reverse characteristics curve of PN junction diode

Apparatus: circuit board, connecting wires.

Circuit diagram: for forward characteristic.



Circuit diagram: for Reverse characteristic



CEDURE FOR V-I CHARACTERISTICS OF PN JUNCTION DIODE:

Forward Bias Characteristics

1. Make all the connections as shown in Fig. (1) using Patch Cord Connect positive socket of Power supply to one end of Anode (A) of PN Junction Diode & also to +ve socket of Voltmeter, Connect Cathode (K) of PN Junction diode to +ve socket of mA meter, Connect -ve socket of Power Supply to -ve of Voltmeter & to -ve of mA meter.
2. Select the power supply range as well as voltmeter range to 2V and current meter to 10 mA range using SPDT switches.
3. Switch on the instrument and set the voltage to 0 volt
4. Increase the voltage in small steps and note down the corresponding current.
Note down the observation in table no.1
5. Plot a graph between voltage and current takin voltage on x axis and current On y axis.

Table.1

Sr. no.	Forward voltage	Forward current
1.		
2.		
3.		
4.		
5.		
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REVERSE BIAS CHARACTERISTICS:

NOTE: Increase the input power supply (0-30V) to achieve only up-to 100 μ A current

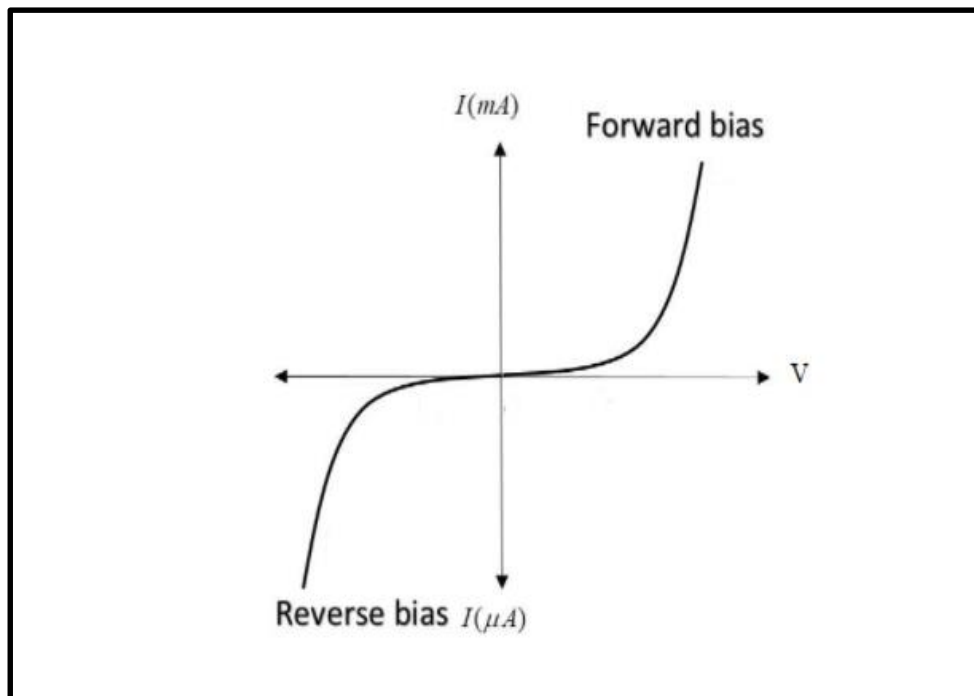
Make all the connection as shown in fig 2 using patch cord connect +ve socket of power supply to one end of resistance R_s , and also to positive terminal of voltmeter. Connect other end of resistance R_S to cathode (K) of diode and anode (A) of diode to positive terminal of μ A meter.

1. Now connect negative terminal of power supply to negative of voltmeter as well as negative terminal of μ A meter
2. Select the power supply range as well as voltmeter range to 30v and current meter to 100 μ A range using SPDT switches.
3. switch ON the instrument & set the voltage to 0 volt
4. Increase the voltage in small steps and note down the corresponding current. Note down the observation in table 2.

Plot a graph between voltage and current for reverse characteristics taking voltage on x axis and current on y axis.

Table 2

Sr. no.	Reverse voltage	Reverse current
1.		
2.		
4.		
5.		
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Graph:**Result:**

Forward characteristic curve of all the diodes are plotted.

Reverse characteristic curves of all the diodes are plotted

Precaution:

1. Connection must be made carefully.
2. Change the value of the voltage in the small increments.